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χ-Space: A Statistical-Spectral Approach for Characterizing Extreme Fluctuation Patterns in Time Series

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The characterization of extreme events (XE) from time series has witnessed an increase of interest due to the great observational occurrence found in several fields, such as space and environmental physics. Motivated by this challenge, we propose a new parameter space (namely, χ-space) composed of two attributes that identify different classes of extreme fluctuations in a time series. Based on reformulated measures for statistical quantiles and singularity spectra, the distance from the origin in the χ-space characterizes the escape from Gaussianity and monofractality that occurs when extreme fluctuations are present in a time series. To generate time series with different patterns of extreme fluctuations, two canonical systems were carefully chosen as illustrative examples: the so-called p-model for multifractal extreme dissipation and the driven Lorenz chaotic model within an appropriate parameterization scheme. The results show that the investigated attributes can compose a two-dimensional space in which the patterns of extreme *endogenous* and *exogenous*

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fluctuations are distinguished with great precision. The characterization of some observed fluctuation patterns from space and environmental physics is presented as a case of practical application.

Keywords: Extreme events; singularity spectrum; non-Gaussian fluctuations; nonlinear dynamics.

1. Introduction

In recent years, the analysis of extreme fluctuations has gained particular attention due to their significant implications for social and natural phenomena (Albeverio *et al.*, 2006; Rosa *et al.*, 2009; Stewart *et al.*, 2022). These patterns, which explain the unusual activity of a system, indicate fluctuations beyond typical ranges, often interpreted as unpredictability in time and non-Gaussianity in the statistical domain. Such fluctuations may also reflect the absence of strategies to mitigate their impact. For example, in the context of a global economic crisis, extreme events can lead to mass layoffs and widespread bankruptcies in an economic sector (Mattedi *et al.*, 2004; Ramos *et al.*, 2001) or geopolitically (Rosa *et al.*, 2019). Similar unusual behaviors are observed in physical, geological, and biological systems, where extreme fluctuations can signal a disruption in equilibrium, potentially resulting in major disruptions, such as mass extinctions. These fluctuations are typically identified as out-of-range deviations in time series such as the number of disease cases (Rezende *et al.*, 2025; Santos *et al.*, 2017; Sautter *et al.*, 2025), ultrasound measurements (Johnson *et al.*, 2021), environmental physics (Campos Velho *et al.*, 2001; Ramos *et al.*, 2004; Rodrigues Neto *et al.*, 2001) and solar activity (Bolzan and Rosa, 2012; Bolzan *et al.*, 2005; Rosa *et al.*, 2008; Veronese *et al.*, 2011), also extending to other domains (Ferreira da Silva *et al.*, 2000; Rempel *et al.*, 2004; Rosa *et al.*, 2018; da Silva *et al.*, 2000), all of which exemplify so-called extreme events (XE). From a practical perspective, the primary objectives of analyzing extreme fluctuations in time series are to understand their underlying causes, identify the energy dissipation processes driving them, and develop strategies for monitoring, forecasting, and controlling their occurrence. However, prediction and control remain long-term challenges due to the complexity and rarity of these events.

Since the formalization of this phenomenological concept, various approaches have been proposed to better understand XE. Despite the diversity of perspectives across different fields, a unified theory of extreme fluctuations is still lacking (McPhillips *et al.*, 2018). The two primary classes of extreme events considered here are *endogenous* and *exogenous* extremes. The distinction between these two classes was introduced by Sornette *et al.* (2004), whose framework differentiates these fluctuations based on the scaling properties of their temporal evolution. In this context, dissipative processes caused by internal instabilities (endogeny) and/or external disturbances (exogeny) produce distinct fluctuation patterns in a time series. A crucial aspect of this approach is that statistical and spectral properties can be leveraged to differentiate between *endogenous* and *exogenous* processes (see Appendix A). From a physical point of view, an XE can be represented as an energy transfer process from low to high frequencies, governed by a probability p, where the energy is distributed in temporal partitions with probabilities p and $(1-p)$ (Meneveau and Sreenivasan, 1991). In a perfectly balanced energy cascade, where $p = 1/2$, the energy is evenly distributed, producing non-extreme signals. In contrast, when the distribution is highly unbalanced, such as in the limiting case of $p = 0$, the energy concentrates in one region at detriment of another, thereby generating an XE. This process is closely related to the multifractal versus monofractal paradigm of analysis, where p determines the distribution properties of the singularity spectrum (see Appendix B). Recent studies have found that the cascading process is not uniformly distributed (Mukherjee *et al.*, 2024), suggesting that no universal law may govern some cascading phenomena.

Understanding the underlying physical processes in extreme events is important, but for a more effective long-term regime characterization, the statistical approach is often preferred. The most popular statistical approach is based on three key distribution attractors: Fréchet, Weibull, and Gumbel, which together form the generalized extreme value (GEV) distribution. GEV usually describes the distribution of the maximum value found in time series segments, which is known as Block-Maxima (BM). When the context requires a larger set of samples, the peak over threshold (POT) approach is used and the generalized Pareto distribution (GPD) is better suited. Both GEV and GPD are characterized by long tails that capture the behavior of XE and deviations from Gaussianity, represented by nonzero third and fourth moments. Although the literature on these distributions is extensive, the connection between their distributional properties and spatiotemporal correlations remains underexplored in many applications. This gap can hinder effective time-series monitoring and forecasting. Moreover, most modern machine learning algorithms such as LSTM, GRU, and Transformers primarily rely on local fluctuations for prediction. Few approaches employ statistical distributions to predict the regime, or set of distribution parameters, instead of local fluctuations in BM or POT distributions (Boulaguiem *et al.*, 2022; Cannon, 2009; Kharin and Zwiers, 2005).

Recent research suggests that statistical approaches, such as GEV and GPD (Boulaguiem *et al.*, 2022; Cannon, 2009; Kharin and Zwiers, 2005), are well suited for capturing the global properties of XE. On the other hand, we observe that local fluctuation deviations are better represented by multifractal and spectral measures. Based on this insight, we propose a new framework, namely, the *χ-space*, which integrates multifractal and statistical metrics. This framework defines a two-dimensional space where the origin (0,0) corresponds to events without extreme fluctuations, typically characterized by Gaussian and monofractal statistical-spectral patterns. The *χ-space* provides a novel approach for analyzing the statistical and spectral properties of time series exhibiting extreme fluctuations.

This paper is structured as follows: In Section 2, we present three models that can produce extreme fluctuations, namely the *p*-model, the Synchronized Lorenz, and the Multifractal Diffusion. The characterization technique, referred to as the χ-space is detailed in Section 3. Our main results, including model analysis and the characterization of real data from space and environmental physics, are presented in Section 4. Finally, we provide concluding remarks in Section 5.

2. Mock Data for Extreme Fluctuations

Our strategy is to identify and use some models that are capable of generating different classes of extreme fluctuations that, to some extent, correspond to the main patterns of XE observed and discussed in the real world. Although the models presented here do not encompass all possibilities, they serve as canonical examples within the proposed methodology. A property common to all is that they are all models that represent dissipative processes since the patterns of endogeny and exogeny (see Appendix A) are related to this property. For each model, a batch with 200 time series was generated and selected. This was an iterative procedure until a representative sample batch was identified for each model.

2.1. *Multiplicative cascade patterns*

Our first intuition is based on a generalization of the so-called *p-model* mechanism (Meneveau and Sreenivasan, 1987, 1991) that allows us to create a non-homogeneous cascade which is entirely compatible with the fluctuations observed in stochastic time series.

The *p-model* approach for a non-homogeneous turbulent-like cascade was proposed by Meneveau and Sreenivasan (1987). It gives a new insight into the kinetic energy dissipation in the cascading process of eddies in the inertial range of a fully developed turbulence, and it is based on the special case of weighted transfer.

From a theoretical point of view, the *p-model* is a generalized form of a two-scale cantor set with a balanced distribution of length, which shows multifractal properties of one-dimensional sections of the dissipation field. The generalized form starts from the classical view of an eddy cascade before the inertial range of a fully developed turbulence, where a flux of energy (E_K) of size L_K cascades down along the inertial range and eventually dissipates when it reaches the Kolmogorov length scale β. Hence, in a cascade step, each eddy of size L_K is divided into two equal parts, $L_K/2$ expressed as L_{K_1} and L_{K_2} each. However, in each cascade step, the flux of energy is distributed, as a probability, unequally in fractions of p_1 and p_2, where $p_2 = 1 - p_1$ and $0 \leq p_2 \leq 1$. This process is iterated over fixed p_1 until the eddies reach the Kolmogorov scale β.

In terms of a multifractal approach, the spatial energy distribution of the *p-model* multiplicative cascade (Meneveau and Sreenivasan, 1987) can be described by

$$\alpha = \frac{\log_2 p_1 + (\omega - 1)\log_2 p_2}{\log_2 l_1 + (\omega - 1)\log_2 l_2} \tag{1}$$

and

$$f(\alpha) = \frac{(\omega - 1)\log_2(\omega - 1) - \omega\log_2\omega}{\log_2 l_1 + (\omega - 1)\log_2 l_2}; \tag{2}$$

where ω is a term related to the Stirling approach for the generalized Cantor set, and $f(\alpha)$ is the multifractal spectrum. For the case of equal-sized eddies ($l_1 = l_2 = 1/2$), ω can be expressed in terms of α, p_1, and p_2:

$$\omega = \frac{\log_2(p_2) - \log_2(p_1)}{\alpha + \log_2(p_2)}. \tag{3}$$

We can further simplify $f(\alpha)$ by expressing it explicitly in terms of ω, which yields

$$f(\alpha) = \frac{\omega\log_2(\omega) - (\omega - 1)\log_2(\omega - 1)}{\omega}. \tag{4}$$

Substituting Equation (3) into expression (4) yields an equation for $f(\alpha)$ which is only dependent on p_1, p_2, and α. We can use this proxy to construct a time series that displays an equivalent multifractal spectrum, but now in the time domain.

Deviations from the homogeneous cascade ($p_1 = p_2 = 0.5$) are compounded by abrupt changes in the frequency and magnitude of the time series. Such changes are called XE. When the level of amplitudes increases significantly due to internal factors, then the extreme events are called endogenous ($\mathrm{XE_{endo}}$). In contrast, it is called an extreme exogenous event ($\mathrm{XE_{exo}}$) when external energy transfer or abrupt dissipation is the main cause of XE. To generate time series $\mathrm{XE_{endo}}$ and $\mathrm{XE_{exo}}$, we will adopt the Sornette *et al.* (2004) formalism (see Appendix B). Typical endogenous and exogenous processes can be obtained with the p-model for $0.525 < p_1 < 0.65$ and $0.675 < p_1 < 0.8$, respectively. Figure 1 shows an example of each class of time series generated with this model.

Although the *p-model* was originally inspired by turbulent eddy dissipation, its connection to classical turbulence metrics remains underexplored. Many applications rely on the Power Spectral Density $S(f)$, defined as

$$S(f_j) \approx \frac{1}{f_j^{\beta}}, \tag{5}$$

where f_j is a given frequency of the signal, and in a fully developed turbulent regime, the scaling exponent approaches Kolmogorov's $\beta = 5/3 \approx 1.667$. However, not all signals following a power-law spectrum exhibit extreme event statistics. This highlights the need for a model that explicitly incorporates the power-law exponent as a tunable parameter, ensuring a more comprehensive description of dynamical behaviors.

2.2. *Diffusive-like patterns*

Colored noise is a widely used model for studying time series, given its prevalence in many practical applications. To generate a signal that follows a specific β colored noise, we employ the squared-root algorithm, similar to the approach proposed by Timmer and Koenig (1995). In Equation (6), the Fourier coefficients

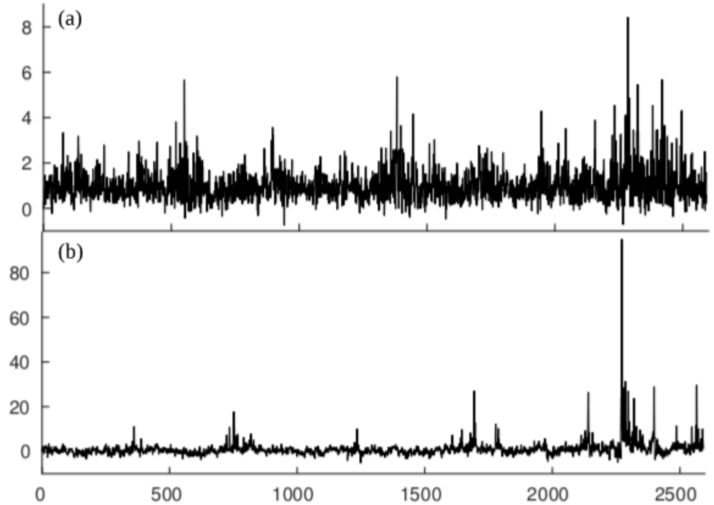


Fig. 1. Typical Time Series for (a) XE_{endo} Series Generated with $p = 0.6$ and (b) XE_{exo} Generated with $p = 0.8$, According to the Sornette *et al.* (2004) Statistical Model

($\hat{x}(f_j)$) are constructed from a standard normal series (η) by normalizing by the respective inverse squared-root power law (β) of each frequency (f_j). To retrieve the colored series, the inverse of the Fourier transform is applied to the synthetic colored noise spectrum, and the imaginary part is discarded:

$$\hat{x}(f_j) = \left(\frac{1}{f_j}\right)^{\frac{\beta}{2}} \eta + i \left(\frac{1}{f_j}\right)^{\frac{\beta}{2}} \eta. \tag{6}$$

In the fractal perspective, every colored noise is associated with a different dissipation law (Figure 2). While white noise signals ($\beta = 0$) are commonly studied, some literature defines their behavior as non-fractal due to the characteristics of their autocorrelation and partial autocorrelation functions (Stadnitski, 2012). Other studies have extended the fractal classification of noises to include monofractal and non-fractal signals (Lovejoy *et al.*, 2000; Márquez-Rámirez *et al.*, 2012). Despite the discussion regarding the mono- or non-fractal behavior of white noise, its significance remains due to its Gaussian distribution and the potential for a singular fractal generative law, which yields the origin of the χ-space.

In contrast to white noise, red noise ($\beta = 2$) signals have a wide range of fractal behavior. We observe that the multifractal spectrum can have monofractal or multifractal characteristics depending on the series trend, which can be described as a combination of spectrum components (Hu *et al.*, 2001). Furthermore, changes in the trend are reflected in the amplitude distribution, which can be effectively modeled using a Gaussian mixture distribution. A key observation is that increasing multifractality in the signal correlates with both an increase in the number of Gaussian components required to model the distribution and a greater separation between these components.

Although the *p-model* serves as a standard framework for turbulence dissipation and the colored-noise as a standard framework in terms of power spectra, alternative mechanisms with distinct dissipative origins have been explored in the literature. In particular, chaotic systems near synchronization have been identified as a novel source of extreme fluctuations (Chowdhury *et al.*, 2021; Frolov and Hramov, 2021) offering a complementary perspective on their emergence and dynamics.

2.3. *Chaotic extreme patterns*

In chaotic systems operating near a synchronization manifold, synchronized orbits exhibit intermittent departures and subsequent re-synchronization. This behavior manifests as on-off intermittency, characterized by a synchronization parameter

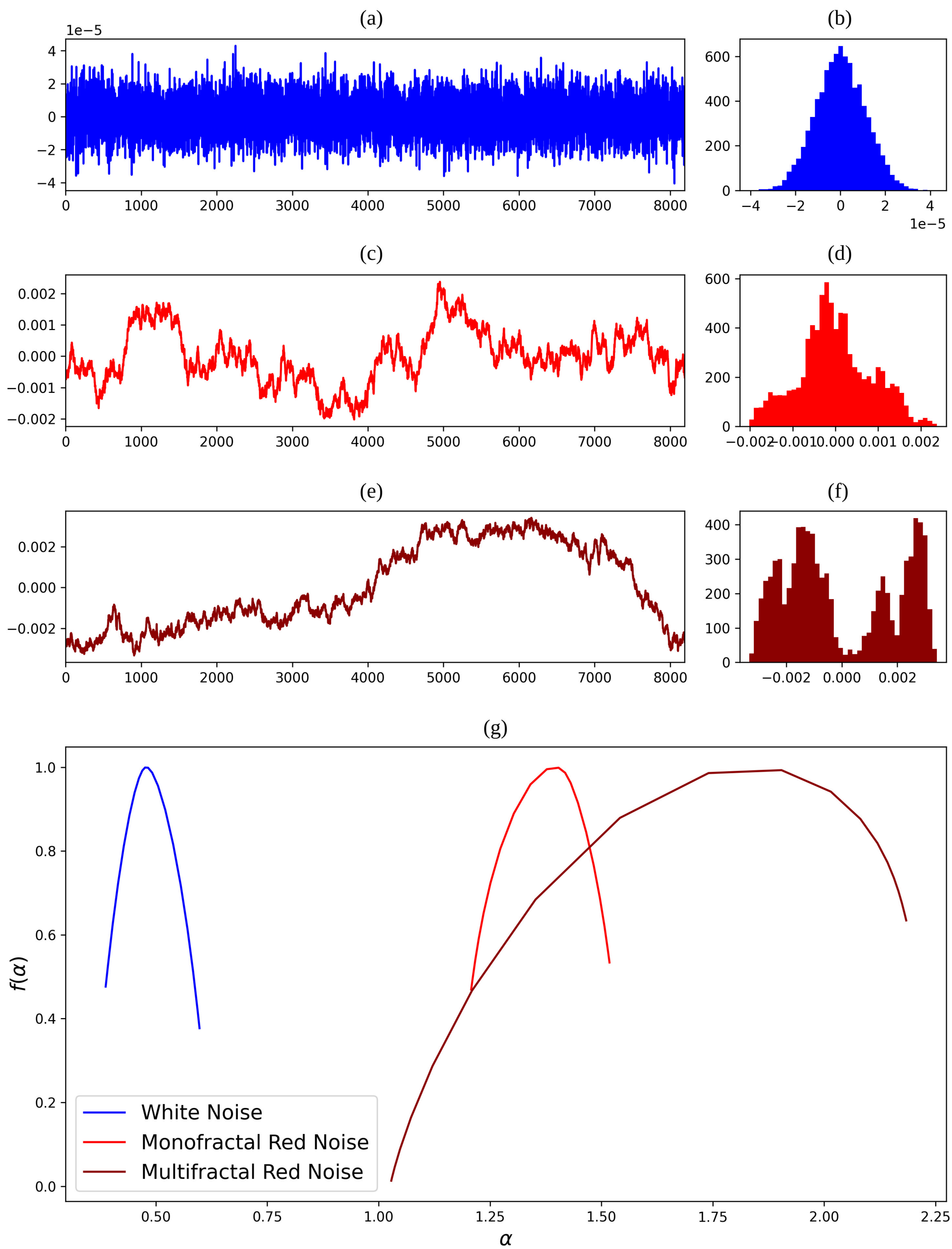


Fig. 2. Colored Noise Multifractal Spectrum Analysis: (a)–(c) Are the Time-Series, (d)–(f) Are the Corresponding Amplitude Distribution, and (g) Is the Multifractal Spectrum of all Time-Series

time series that fluctuates between near-zero values during synchronization and nonzero values during de-synchronization (Chowdhury *et al.*, 2021).

In this study, we investigate a synchronized version of the Lorenz system, which can generate chaotic extreme patterns within specific parameter regimes. Building upon the classical Lorenz model, we propose the following system of coupled differential equations:

$$\begin{aligned} \frac{d}{dt}x_\mu &= \sigma_\mu(y_\mu - x_\mu) - \mu(x_\mu - x_0), \\ \frac{d}{dt}y_\mu &= x(\rho_\mu - z_\mu) - y_\mu - \mu(y_\mu - y_0), \\ \frac{d}{dt}z_\mu &= x_\mu y_\mu - \beta_\mu z_\mu - \mu(z_\mu - z_0). \end{aligned} \tag{7}$$

Note that we first integrate the Lorenz system for $\mu = \mu_0 = 0$ and parameters $\sigma_0 = 10$, $\beta_0 = 8.3$, and $\rho_0 = 28$ to obtain the undriven solution to the Lorenz system $\{x_0(t), y_0(t), z_0(t)\}$. We use this signal to drive the Lorenz system with $\mu \neq 0$ and parameters $\sigma_\mu = 10$, $\beta_\mu = 8.3$, and $\rho_\mu = 20$ to obtain the driven solution to the Lorenz system $\{x_\mu(t), y_\mu(t), z_\mu(t)\}$.

The reader may note that the terms $\mu(x_\mu - x_0)$, $\mu(y_\mu - y_0)$, and $\mu(z_\mu - z_0)$ work as a driven term of the system, and the parameter μ defines the level of the synchronization. This study utilizes the parameter sets detailed in Table 1 for simulations of the Lorenz System (LS) and the Driven Lorenz System (DLS).

Metrics like the Conditional Lyapunov Exponent (Shuai *et al.*, 1997) and Topological Entropy (Caneco *et al.*, 2009) are robust to detect synchronization. However, their pitfall is exactly the robustness in this application, since their inherent focus on long-term behavior diminishes sensitivity to transient local extremes. Instead, we employ the Euclidean distance $d_\mu(t)$, which effectively captures abrupt state changes, aligning with the approach of Chowdhury *et al.* (2021) for detecting extremes in chaotic oscillators, namely

Table 1. Lorenz System and Driven Lorenz System Parameters. The Values for the Coupling Term (μ) in Driven Lorenz System Is a Random Realization Between 2 and 5.

Parameter	Lorenz System	Driven Lorenz System
σ	10	10
β	8/3	8/3
ρ	28	29
μ	0	[2–5]

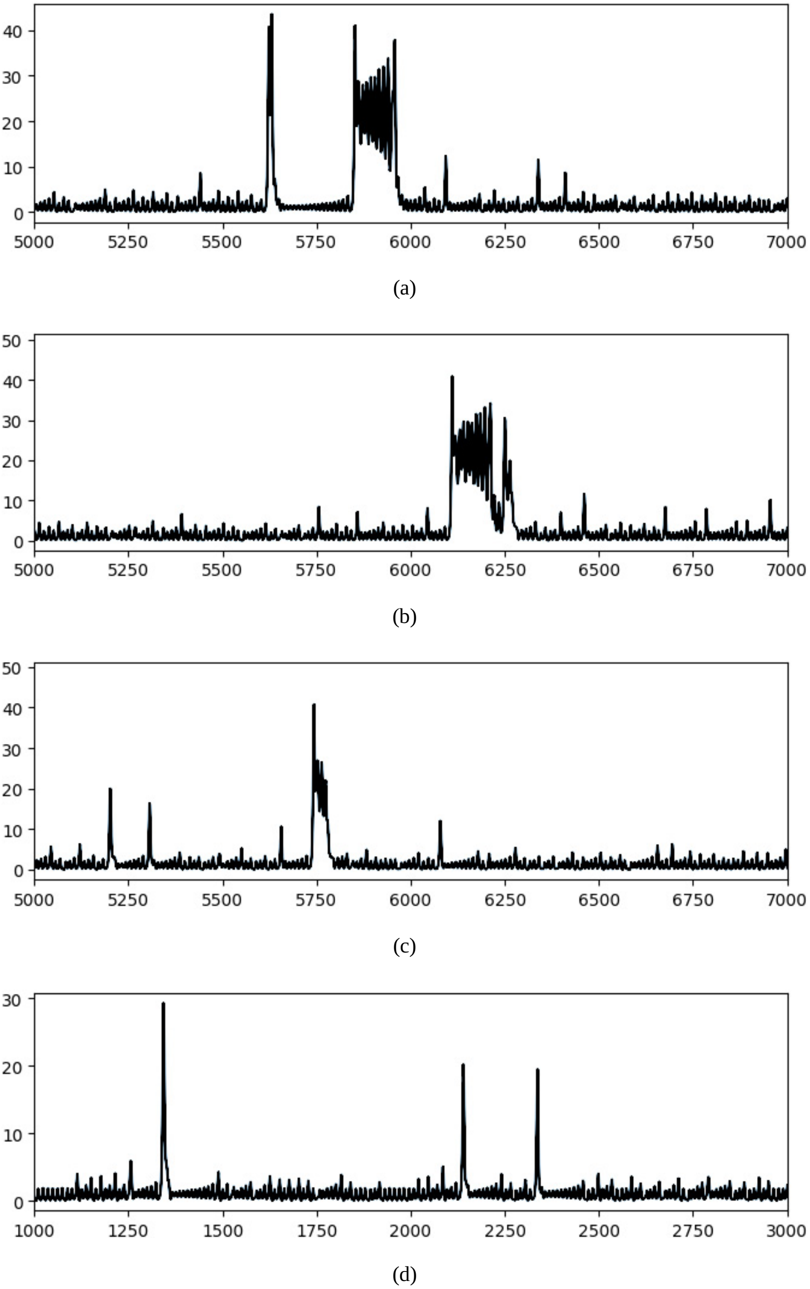


Fig. 3. Four Samples of Chaotic-Synchronization Extremes. The Extreme Patterns Observed in (a), (b), and (c) Exemplify On–Off Intermittency, Characterized by Intermittent Bursts of Desynchronization Followed by Resynchronization at Different Levels. In Contrast, (d) Shows an Extreme Time Series with a More Continuous Synchronization Dynamics, Where the Extremes Emerge Without the Intermittent Switching Typical of On–Off Intermittency.

$$d_\mu(t) = \sqrt{(x_\mu(t) - x_0(t))^2 + (y_\mu(t) - y_0(t))^2 + (z_\mu(t) - z_0(t))^2}. \tag{8}$$

Analyzing the evolution of $d_\mu(t)$ allows us to discern between synchronized and locally desynchronized states within the system's orbits. This time series of Euclidean distances constitutes a novel form of extreme event within our simulated dataset. Notably, varying the parameter μ yields diverse dynamical behaviors, as illustrated in Figure 3.

3. The χ Parametric Space

To quantify and compare the diversity of extreme time series, we propose a parametric space χ as a composition of a distribution metric and a multifractal scaling metric.[1] The components of χ-space are the Gaussian Quantile–Quantile plot distance (G_{QQ}) and the normalized width of the singularity spectrum $\Delta\alpha_N$. The G_{QQ} is a normalized measure of the Gaussian distribution fitting error, whereas $\Delta\alpha_N$ indicates the non-fractal, monofractal, or multifractal composition of the signal.

The Quantile–Quantile plot (QQ plot) is a visual technique that aims to determine whether a population can be described by a given distribution function (Figure 4). In this technique, the population quantiles and the corresponding quantiles of the hypothetical distribution are plotted in separated axis. If the population has the same distribution as the theoretical model, then the graph will follow a linear rule. In the case of the presented technique, our goal is to detect extreme fluctuations which deviates from the long-term dynamical range. Thus, the QQ plot of the normalized data population, with zero average and standard deviation of 1, is compared with a standard normal distribution. We propose a metric G_{QQ}, which is the inverse of the coefficient of determination. Equation (9) shows the simplification of the inverse of the coefficient of determination, where P_Q is the normalized population quantile Q, η_Q is the quantile Q of the standard normal distribution, and $\overline{\eta_Q}$ is the average of all standard normal distribution quantiles

$$G_{QQ} = \frac{\sum (P_Q - \eta_Q)^2}{\sum (\eta_Q - \overline{\eta_Q})^2}. \tag{9}$$

For the second metric, we extended the Multifractal Detrended Fluctuation Analysis (MFDFA) technique proposed by Gorjão *et al.* (2022). The traditional

[1]https://github.com/rsautter/Chi

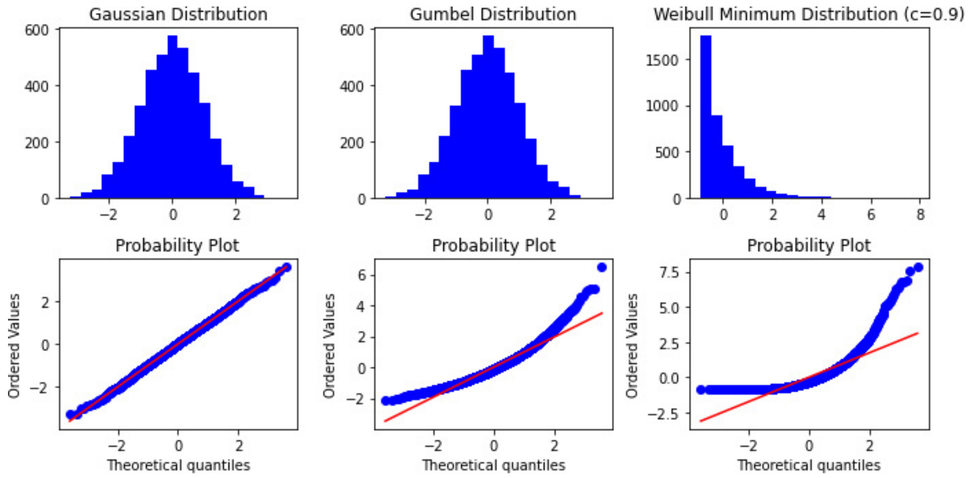


Fig. 4. (Color online) *QQ* Plot Analysis of Three Distributions: Normal Distribution, Gumbel Distribution, and Weibull Minimum Distribution. The Red Line Indicates the Best Fit of the Data

MFDFA is a semi-supervised method, which requires a user to determine the best scales of analysis and the order of Hurst exponent fluctuation (q). Some applications demand a higher level of automation, which can be an issue for this approach. Also, the choice of the set of qth exponents depends on the fractal nature of the signal, which may require some expertise of the user. We have observed that setups that do not match with the nature of the signal usually have a greater variation of singularity metrics (Appendix B), thus removing the outliers can approximate the best setup. We propose the following algorithm to automate the multifractal spectrum analysis:

(1) Normalize the time series by its average and standard deviation.
(2) Measure the singularity spectrum with a set for the qth order range, with monofractal and multifractal q exponents.
(3) Compute the Box-asymmetry of every singularity spectrum (see Appendix B).
(4) Remove half singularity spectra set with higher Box-asymmetry.
(5) Fit the quadratic equation $f_\alpha(\alpha) = x_1\alpha^2 + x_2\alpha + x_3$ of the singularity spectrum, and find the roots α_1 and α_2 to define $\Delta_\alpha = |\alpha_2 - \alpha_1|$.
(6) Normalize $\Delta\alpha$ according to a polynomial function [Equation (10)]

Step 1 is important to enhance the fluctuations of the time series and avoid divergences in the analysis of signals from different sources. The following steps 2–4 define the best spectrum measures, which are defined here by the lowest box-asymmetry measure, which prioritize singularity spectra with low variations of

$f(\alpha)$ at α extremes, and monofractal singularity spectrum. Step 5 addresses the size dependency of $\Delta\alpha$, facilitating comparisons across time series of varying lengths. Step 6 normalizes $\Delta\alpha$ using the following polynomial function:

$$\Delta\alpha_N = \frac{\Delta\alpha}{\Delta\alpha + 1}. \tag{10}$$

Equation (10) ensures that $0 \leq \Delta\alpha_N \leq 1$, since any distance metric for the singularity spectrum is non-negative. This automated framework provides a robust method for characterizing multifractal signals. In the subsequent section, we evaluate the performance of the proposed metric using synthetic data and explore the relationship between this statistical metric and physical–statistical metrics.

4. Results and Interpretation

Finding the most suitable model for a given observational dataset is a complex task. The Cullen–Frey space (Cullen *et al.*, 1999; Lazoglou *et al.*, 2019) is a popular approach to this end, where the third and fourth statistical moments are used to identify the optimal distribution. Similarly, we introduce a mock dataset of extreme events in the χ-space. Several models can generate extreme time series, as previously shown in this work, with slight variations related to statistical distribution and fluctuations. In this sense, the proposed metrics aim to enhance the classical technique by introducing a structural metric and a more generic parameter in the statistical paradigm.

4.1. *Mock data*

The mock data are a set of 200 time series samples of each of the systems described in Section 2, resulting in the χ-space depicted in Figure 5. This representation reveals distinct characteristics of the mock data and different regions that each type of system seems to occupy.

The standard P-model is aligned with the main diagonal of the χ-space, where Exogenous series occupy the region $G_{QQ} > 0.5$ and $\Delta\alpha_N > 0.5$. This indicates the main property of the model: the extreme statistics and the spatiotemporal distribution are linearly proportional to the cascading imbalance.

In contrast, colored noise series show minimal variations in their statistical distribution, despite exhibiting varying cascading processes. These series, characterized by Gaussian or Gaussian-composite distributions, deviate significantly from the extreme distribution perspective, resulting in $G_{QQ} \approx 0.0$. While red noise

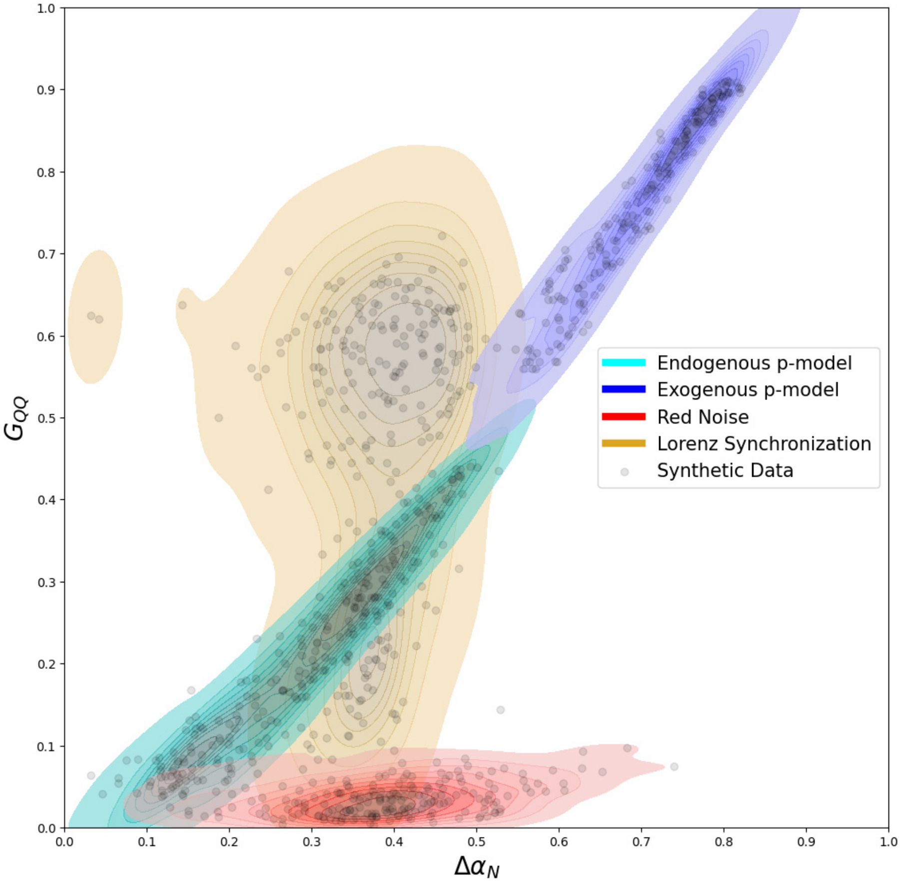


Fig. 5. χ-Space for Mock Data. Each Dot in the Figure Corresponds to the Location of a Mock Data Sample in the χ-Space

amplitudes may not exhibit extremes, the observed cascading patterns still indicate substantial dissipation, crucial to understanding potential system changes. In contrast to colored-noise series, the set of chaotic-synchronization series are almost invariant to the cascading distribution, which are endogenous ($\Delta\alpha_N = 0.35 \pm 0.15$) in this perspective.

An exception to this cascading invariance occurs in synchronized series with constant distance, as illustrated in Figures 3(a) and 3(b). These cases, characterized by synchronization parameters slightly exceeding two ($\mu \approx 2 + 10^{-5}$), represent transitional states between synchronization levels.

These properties enable the characterization of extreme time series and the selection of models that closely resemble the observed phenomena. Furthermore, given the segregation of the different types of mock data in different regions of the χ-space, this approach allows for the identification of the underlying mechanisms driving extreme events. By mapping real-world data to the χ-space, we can gain insights into the system's inherent dynamics and assess its vulnerability to extreme

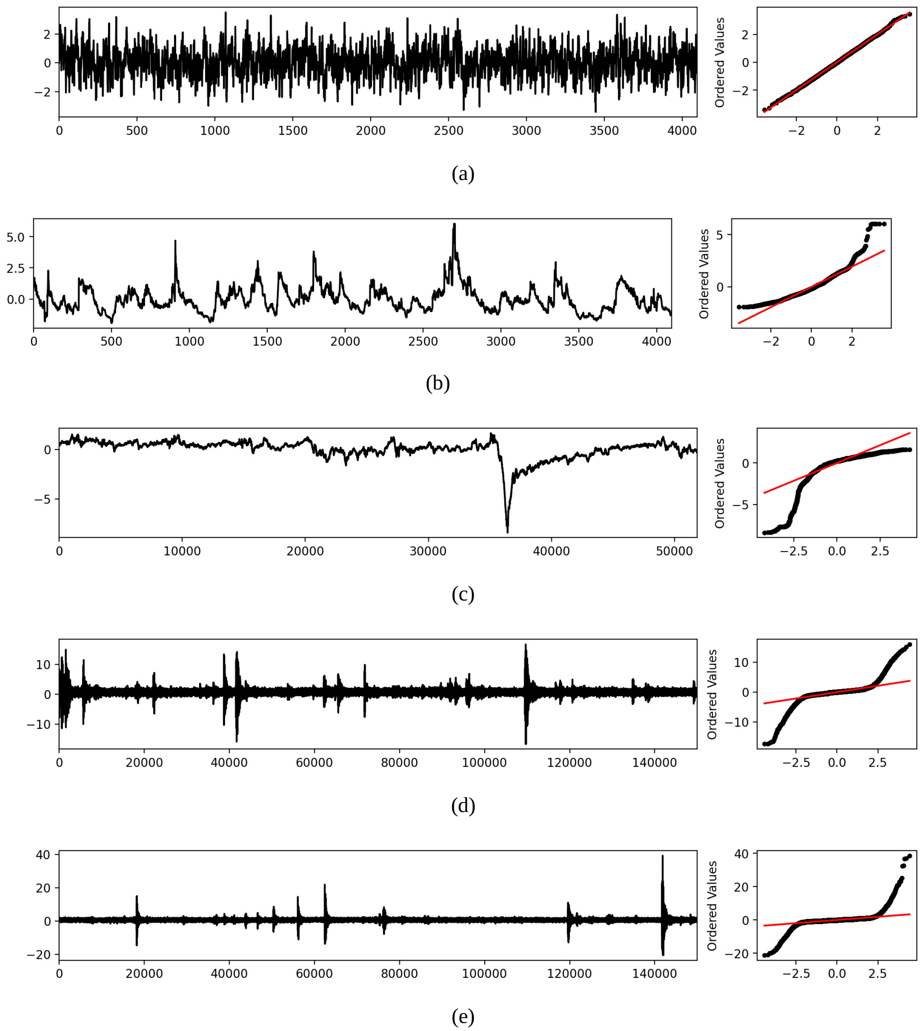
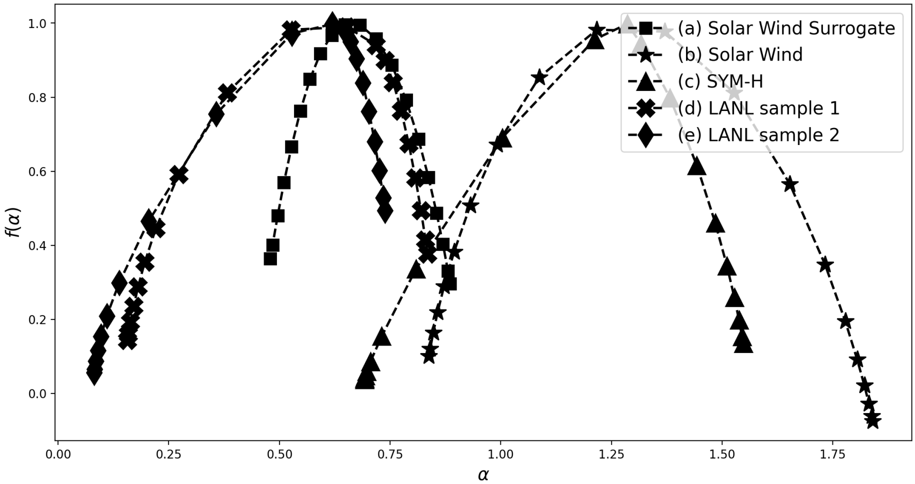


Fig. 6. Collection of Extreme Real-World Data Samples and Respective *QQ* Plot. Panels (a) and (b) Depict Surrogate and Original Time-Series of Solar Wind Data (Rosa *et al.*, 2008), Respectively, Illustrating the Tendency Toward a Gaussian-like Pattern After Applying the Surrogate Transformation. Panel (c) Presents the SYM-H Geomagnetic Index from August to September 2018 (Alberti *et al.*, 2020), While Panels (d) and (e) Display Ultrasonic Observations from Simulated Earthquake Experiments (Johnson *et al.*, 2021). Finally, Panel (f) Shows the Singularity Spectrum for All Real Data Samples



(f)

Fig. 6. (*Continued*)

fluctuations. This methodology offers a powerful tool for comparative analysis, enabling the systematic evaluation of various systems exhibiting extreme behavior.

4.2. *Real data*

To demonstrate practical applications, we present the use of the χ-space framework in space weather and seismology. Extreme events in these fields carry profound social and economic consequences, highlighting the critical need for further research. Figure 6 showcases representative datasets, including: 3 GHz solar wind observations (Rosa *et al.*, 2008), the SYM-H geomagnetic index (Alberti *et al.*, 2020), and ultrasonic vibrations preceding a simulated earthquake (Johnson *et al.*, 2021).

The solar radio observation in Figure 6(a) does not exhibit extreme values. However, it corresponds to an active sunspot region that has been continuously monitored. Notably, despite having a low G_{QQ}, this series exhibits a high $\Delta\alpha_N$ value, consistent with the behavior observed in many red-noise signals with a Gaussian-composite distribution. By applying a surrogate transformation, in which amplitudes of the time series are shuffled, the cascading process is disrupted with minor changes in G_{QQ} metric. Both the SYM-H index and the simulated earthquake datasets exhibit endogenous properties, with the earthquake series showing a notable deviation toward chaotic synchronization (Figure 7).

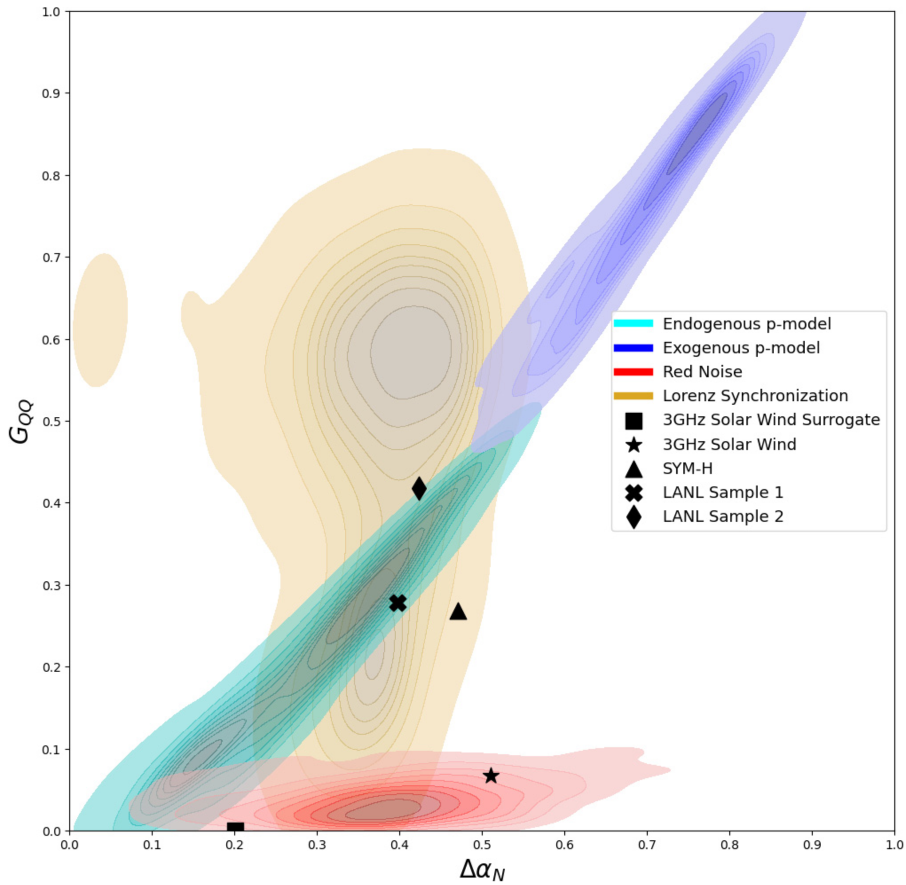


Fig. 7. Projection of Real Data in χ-Space. Each Point Represents a Single Observation, and the Color of Each Point Indicates Its Corresponding Mock Datatype

5. Concluding Remarks

This study proposed χ-*space*, a novel parameter space for characterizing extreme events in time series by leveraging reformulated measures for statistical quantiles and singularity spectra. Analysis of canonical systems demonstrates the χ-*space*'s efficacy in distinguishing between endogenous and exogenous extreme fluctuations. Importantly, χ-*space* reveals potential divergences between statistical and energy cascade perspectives on extreme events, highlighting the need for further investigation into the physical interpretations of these discrepancies. For instance, a red-noise signal, typically Gaussian or Gaussian-composition distributions, can exhibit varying extreme and non-extreme cascading processes. Conversely, the observed extreme values, of chaotic origins, show a constant cascading process, despite the distribution. Alongside theoretical models, the successful application of χ-*space* to real-world data from space and environmental physics showcases its potential to advance our understanding and of prediction of extreme events across diverse scientific disciplines. This methodology offers a promising new avenue for analyzing complex systems and extracting valuable insights from time series data. Based on these findings, future work aims to extend this methodology to the characterization of extreme events in two dimensions, such as reaction-diffusion systems and an expanded version of the 2D *P*-model, which adapts the presented approach to bidimensional domains.

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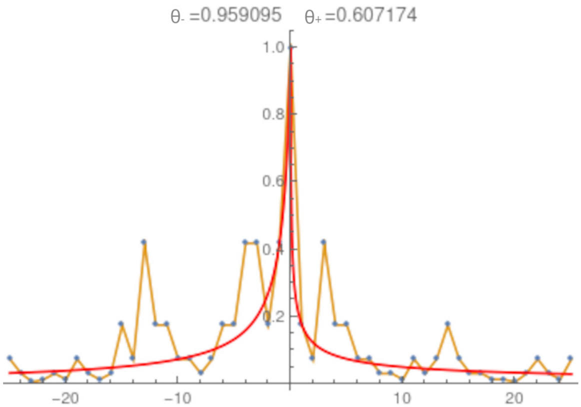
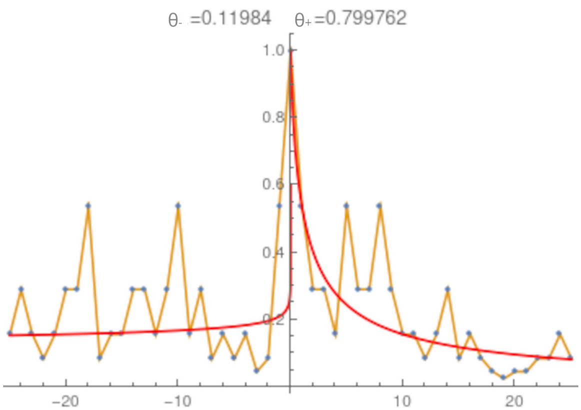
Appendix A. Endogenous and Exogenous Time Series

Extreme fluctuations are prevalent in various natural processes. From a criticality standpoint, it is expected that minor fluctuations precede every event. However, practical limitations often prevent the detailed observation of events at a high resolution. Certain events occur rapidly, exceeding the capacity for direct observation, thus making their prediction more challenging. In this regard, the concept of endogenous–exogenous events has been introduced in the literature Sornette (2006).

In the framework of extreme events, this study has progressed in recent years to the point where we can identify a statistical model for the (XE_{endo}) and exogenous (XE_{exo}) time series. We will adopt here the model proposed by Sornette *et al.* (2004). Following the approach proposed by Sornette *et al.* (2004), internal perturbations give rise to XE_{endo} which is characterized, as shown in Figure A(a), by smoother continuous fluctuations on both sides of the peak that increases slowly and after reaching its highest peak, gradually reduces by itself. On the other hand, XE_{exo} results from an external perturbation and can be characterized by a sudden peak followed by an unexpected rapid drop in the fluctuations [Figure 8(b)]. Around the peak, less continuous fluctuations can be observed.



(a)



(b)

Fig. A.1. Typical Time Series X_E Clipped Round Its Extreme, Where (a) is a XE_{endo} Series and (b) is a XE_{exo} Series. The Points on the Left Size of the Peak of the Series Were Fitted with the Sornette *et al.* (2004) Model Parameter θ_-, while for the Points on the Right Size of the Peak with Parameter θ_+

The Sornette *et al.* (2004) model is based on the book sales rank where the higher the sale for any book, the lower its sales rank (Figure A.1). Among thousands of books, they select the sales ranking of two prototype books. While the book's selling rate, which has a XE_{endo} pattern, only relies on the advertising provided by the common sales system (basically, the publisher's advertising and, especially, the cascade of information between the readers and likely readers), the sales rate of the book with a XE_{exo} pattern accounted on an unusual high-cost advertisement via a famous newspaper. The XE_{endo} time series around the peak was found to be nearly symmetrical, with smoother fluctuations, and took almost four months for the whole process. The appearance of the peak is a result of internal dynamics of system without any influence of external factors.

According to the Sornette *et al.* (2004) approach, time series are modeled based on *social epidemic process* where in the beginning, first (*mother*) agent notices the book in advertisement or news or by chance and buy it at time t_i. Subsequent (*daughter*) generations of agents are built at different time t resulting in an epidemic that can be modeled by a memory kernel $\phi(t - t_i)$. The net sale is the sum of white-noise processes following a power-law distribution that accounts for $\mathrm{XE}_{\mathrm{endo}}$, and *semi-Dirac* distribution associated with $\mathrm{XE}_{\mathrm{exo}}$. The time series can be described by a conditional Poisson branching process given by

$$\lambda(t) = R(t) + \sum_{i/_1 \leq t} \mu_i \phi(t - t_i), \tag{A.1}$$

where μ_i is number of potential agents influenced by the agent i who bought earlier at time t_i. $R(t)$ is the rate of sales initiated spontaneously without influence from other previous agents. For our generic complex systems scenario, the key idea in the Sornette *et al.* (2004) model, is the invariance of the epidemic model but as an inhomogeneous network of potential daughter generations which can be considered through different values of branching ratio.

The ensemble average yields a *branching ratio*, n, that signifies the average number of conflicts triggered by any *mother Agent* within her contact network and relies upon the network topology and impact of the geopolitical behavior. The authors considered the sub-critical regime $n < 1$ in order to ensure stationarity which accounts for efficient coarse-grained nature of the complex nonlinear dynamics. The exogenous response function is obtained from the Laplace transform of the Green function $K(t)$ of the ensemble average.

According to Sornette *et al.* (2004) a *bare propagator* as $\phi(t - t_i) \sim 1/t^{(1+\theta)}$ with $0 < \theta < 1$ corresponds to long-range memory process which provides information on the conflicts propagation:

$$C_{\mathrm{exo}}(t) \equiv K(t) \sim 1/(t - t_c)^{1-\theta}. \tag{A.2}$$

It provides information about the average number of agents influenced by one agent through any possible direct descent or ancestry. And, therefore the average number of conflicts triggered by one agent can be given as

$$\int_0^{\infty} K(t)dt = n/(1 - n). \tag{A.3}$$

Continuous stochastic time series with spontaneous peaks indicate the lack of exogenous shock. Such series can be interpreted as an interaction between external factors over small-scale and enlarged effect of a widespread cascade of social influences. This mechanism can explain the peak in endogenous time series. Considering results for stochastic processes with finite variance and covariance for average growth of processes prior and later to the peak and applying to $\lambda(t)$ defined in Equation (A.1), one gets

$$C_{\text{endo}}(t) \sim 1/|t - t_c|^{1-2\theta}. \tag{A.4}$$

Equations (A.2) and (A.4) agree with the prediction that XE_{exo} should occur faster with exponent $1 - \theta$ compared to XE_{endo} with exponent $1 - 2\theta$.

Appendix B. Singularity Spectrum Metrics

Our library has the following singularity spectrum metrics implemented: $\Delta\alpha$, $\Delta f(\alpha)$, $A_{\Delta\alpha}$, and $A_{f(\alpha)}$. A scheme of the singularity spectrum metrics is presented in Figure B.1.

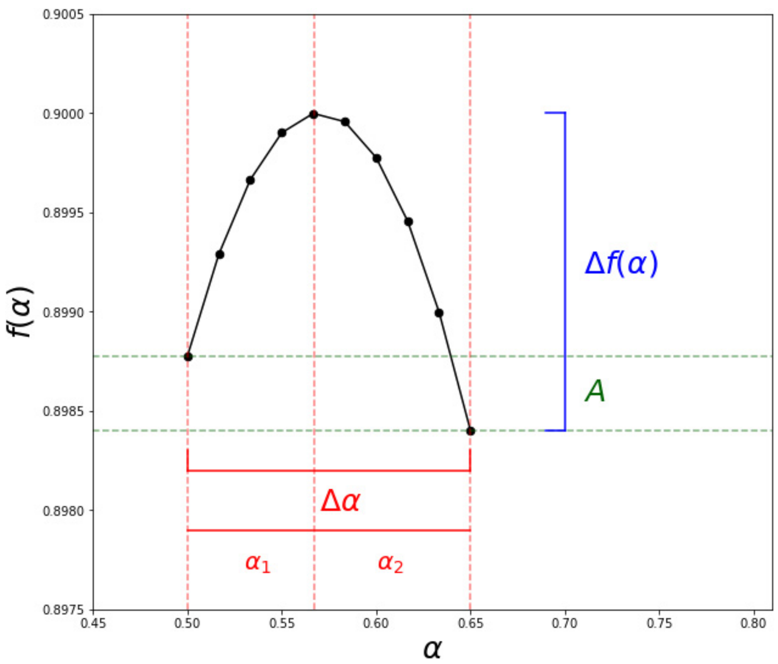


Fig. B.1. Singularity Spectrum Outlining the Typical Quantities Used to Characterize the Degree of Multifractality of the Underlying Process Generating a Time Series. For the Monofractal Case, Both $\Delta f(\alpha)$ and $\Delta\alpha$ Tends to Zero

We propose the box-asymmetry (B_A) as a metric that is derived from the asymmetry (A) and $\Delta\alpha$, which is computed as: $B_A = A\Delta\alpha$.

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